

Mitra: A Multilingual LLM-Based Intelligent Companion and Caregiver Monitoring System for Elderly Assistance

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Abstract—The rapid global demographic shift toward an aging population has exacerbated the prevalence of social isolation, cognitive decline, and inadequate routine management among the elderly. While technological interventions such as digital companions have emerged to address these challenges, existing systems predominantly operate under monolingual constraints, alienating linguistically diverse populations and limiting their widespread efficacy. This paper introduces Mitra, a novel multilingual conversational assistant powered by the Google Gemini Large Language Model (LLM) API and integrated with a comprehensive caregiver monitoring system. Mitra addresses the critical need for native-language accessibility by employing a dual-layer localization architecture: a Flutter-based client-side extension managing Application Resource Bundle (.arb) files for dynamic User Interface (UI) localization, coupled with a language-adaptive prompt engineering framework. The system backend leverages Firebase Cloud Firestore for real-time data persistence and Firebase Cloud Functions for securely executing chatbot workflows, scheduling medication reminders, and processing cognitive brain games. Furthermore, the system incorporates a behavioral analytics dashboard that provides caregivers with quantitative insights derived from an iteratively calculated engagement score. Our proposed architecture establishes the technical feasibility of integrating advanced LLM APIs with proactive caregiver analytics, presenting a scalable solution for next-generation geriatric assistive technologies.

Keywords—Multilingual Conversational AI, Large Language Models (LLMs), Elderly Assistance Systems, Caregiver Monitoring, Assistive Technologies, Engagement Analytics, Human–Computer Interaction (HCI), Digital Health

I. INTRODUCTION

The contemporary demographic landscape is defined by an unprecedented increase in the aging population. According to global health projections, the proportion of individuals aged 65 and over is accelerating, bringing forth complex socio-medical challenges including emotional isolation, medication non-adherence, and cognitive degradation. To mitigate these issues, digital companionship systems have increasingly been deployed. However, a significant barrier to the adoption of these assistive technologies is linguistic accessibility. Older adults, particularly in culturally and linguistically diverse regions such as the Indian subcontinent,

exhibit a strong preference for communicating in their native languages. Monolingual virtual assistants inherently fail to provide the empathetic resonance and cognitive comfort required for effective elderly care.

Furthermore, existing conversational agents often operate as isolated entities, lacking the structural integration necessary to keep medical professionals or family members informed. Effective geriatric care necessitates a hybrid approach wherein the autonomy and emotional well-being of the primary user are prioritized, while simultaneously providing caregivers with transparent, data-driven insights into the user's daily routines and psychological state.

To address these critical gaps, we propose Mitra, an end-to-end ecosystem integrating a multilingual LLM-driven companion bot utilizing the Gemini API and an analytical caregiver dashboard. Mitra is designed to continuously assess behavioral engagement, detect linguistic nuances, and dynamically adapt its computational processing to the user's preferred language natively.

The explicit contributions of this paper are as follows:

Multilingual LLM-based elderly assistant: An end-to-end architecture leveraging the Gemini API and Flutter localization extensions for native-language interaction, minimizing translation latency.

Language-adaptive prompt engineering framework: A robust mechanism for injecting medical constraints, long-term contextual memory, and culturally appropriate conversational guardrails dynamically based on the detected language.

Cloud-native feature execution: Utilization of serverless Firebase Functions to manage the AI conversational workflow, cognitive game logic, and automated medication reminders securely.

Caregiver monitoring dashboard: A secure, Role-Based Access Control (RBAC) interface that aggregates behavioral and physiological data into quantifiable metrics.

II. RELATED WORK

The foundational architecture of modern conversational agents relies on the Transformer architecture introduced by Vaswani et al. (2017), which utilizes self-attention mechanisms to capture long-range dependencies in text. The subsequent scaling of these architectures into Large Language Models (LLMs), as demonstrated by Brown et al. (2020), has enabled few-shot learning and zero-shot contextual adaptations, making open-domain conversational AI highly feasible.

In the domain of multilingual LLM research, significant progress has been made in cross-lingual transfer learning. However, many commercial systems still rely on English-centric pre-training, leading to tokenization inefficiencies and latency issues when processing non-Latin scripts. Current research in conversational AI for healthcare emphasizes the need for systems that not only respond accurately but exhibit empathetic resonance. Studies illustrate that while digital companions improve social interaction, they frequently suffer from short-term memory constraints and rigid, command-based interfaces that frustrate older adults.

Elderly-care assistive systems have traditionally focused on rigid utility—such as scheduled pill dispensers or basic SOS buttons. Modern digital health literature heavily emphasizes the necessity of engagement analytics and reminder compliance tracking systems. However, a critical gap exists at the intersection of these domains. While robust monitoring tools exist, they are rarely coupled with advanced, empathetic, multilingual conversational agents. Mitra bridges this divide by embedding engagement tracking and compliance analytics directly into the natural language interactions of the user, thereby providing a holistic, linguistically inclusive caregiving ecosystem.

III. SYSTEM REQUIREMENTS

The engineering of the Mitra system necessitates strict adherence to both functional and non-functional requirements to ensure reliability, safety, and operational efficiency.

A. Functional Requirements

Multilingual Conversation: The system must seamlessly process, comprehend, and generate natural language text and speech in multiple languages without manual reconfiguration.

Automatic Language Detection: The system must autonomously identify the input language from textual or transcribed audio streams.

Language-Consistent Reminder Generation: All system-initiated prompts, including medication reminders and wellness check-ins, must be delivered in the currently active

language of the elder.

Caregiver Dashboard: A centralized portal must aggregate compliance logs, health vitals, and algorithmic engagement metrics.

Engagement Tracking: The system must quantitatively track conversational frequency, interaction duration, and routine compliance to calculate a rolling baseline of the user's operational state.

Alert System: The architecture must include an SOS module capable of overriding conversational processing to dispatch immediate alerts to registered caregivers.

B. Non-Functional Requirements

Latency Thresholds: To maintain conversational naturalness, the end-to-end response latency must not exceed 2.0 seconds under standard network conditions.

Secure Storage: All Personally Identifiable Information (PII) must be encrypted at rest and in transit.

Role-Based Access Control (RBAC): Cryptographic separation must be maintained between elder and caregiver roles, ensuring caregivers cannot access raw conversational transcripts.

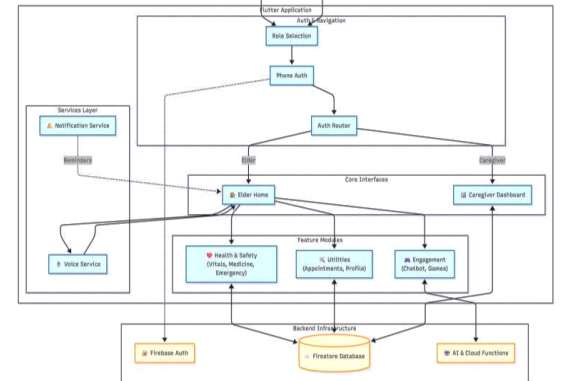
Accessibility Compliance: The graphical user interface must feature high-contrast color palettes, large typography, and voice-first interaction pathways.

Scalable Cloud Architecture: The backend infrastructure must support horizontal scaling and serverless execution.

IV. SYSTEM ARCHITECTURE

The Mitra architecture is a distributed, multi-tier system composed of a client-side Flutter application, a serverless Firebase backend, and cloud-native integration with the Gemini API.

Fig 1: Overall System Architecture Diagram
Architecture Diagram



A. Architectural Modules

Elderly Client Interface: Developed utilizing the Flutter framework. This layer handles local state management, Speech-to-Text (STT), and Text-to-Speech (TTS). It strictly manages static UI localization via a Flutter multilingual extension.

Backend REST API: Implemented via Firebase Cloud Functions to securely execute business logic, manage API calls, and hide the Gemini API keys from the client-side application.

Language Detection Module: A middleware component that evaluates the incoming UTF-8 string to determine the script and linguistic origin.

Prompt Engineering Module: Dynamically constructs the system prompt by appending language rules, the user's memory summary, and strict behavioral guardrails.

LLM Integration Layer: Interacts directly with the Google Gemini API, handling payload formatting and parameter tuning (e.g., temperature adjustments for empathetic conversational variance).

Database Layer: Cloud Firestore serves as the primary NoSQL document database, structured with distinct collections for users, chat logs, vitals telemetry, and caregiver mappings.

Caregiver Dashboard: A responsive web-based application querying real-time analytical views, focusing entirely on metadata and aggregated engagement scoring.

B. Chatbot Implementation Workflow

The system employs a specific, secure workflow to route user queries through the Gemini API while maintaining multilingual context:

User Input Phase: The user speaks into the device. The STT engine transcribes the audio and passes the localized string via an HTTP POST request to a dedicated Firebase Cloud Function.

Context Retrieval: The Cloud Function parses the string and concurrently fetches the memorySummary and recent conversationHistory documents from Firestore to maintain cross-session memory.

Prompt Construction: The system selects a meta-prompt structure corresponding to the detected language and injects the retrieved context.

Gemini API Inference: The Cloud Function securely invokes the Gemini API endpoint. The model processes the unified prompt and generates an empathetic, culturally congruent

response natively in the target language.

Response Delivery & Logging: The text is returned to the Flutter client for TTS synthesis. Concurrently, the Cloud Function logs the new exchange into Firestore and asynchronously updates the engagement analytics engine.

V. DESIGN AND IMPLEMENTATION

A. Firebase Backend and Feature Execution

Mitra heavily relies on Firebase's serverless ecosystem to offload complex processing from the mobile client, ensuring a lightweight and responsive application.

Cloud Firestore: Serves as the real-time synchronization engine. When a user logs a vital sign (e.g., Blood Pressure) or completes a cognitive game, the data is instantly written to Firestore, which immediately triggers snapshot listeners on the Caregiver Dashboard.

Firebase Cloud Functions: Acts as the centralized execution environment for advanced features:

Chatbot Orchestration: As detailed in Section IV.B, Functions securely handle all Gemini API communications.

Reminders & Alerts: Scheduled cron jobs (via Google Cloud Scheduler paired with Firebase Functions) routinely check the medications sub-collection. When a reminder is due, the Function dispatches a localized push notification to the elder's device.

Brain Games Logic: Cognitive games require state validation. Cloud Functions validate high scores, process Sudoku grid generations, and calculate cognitive interaction durations to feed into the user's Engagement Score.

B. Multilingual Support Mechanism

Effective multilingual support requires natively semantic processing. Mitra achieves this by combining a Flutter multi-lingual extension (utilizing .arb files or similar localization dictionaries) for static UI elements with the natively multilingual Gemini API for dynamic conversation.

The client-side Flutter extension reduces computational overhead by handling all deterministic translations locally (e.g., "Settings", "Emergency", "Medication"). When conversational input is detected, the Prompt Engineering Module selects a corresponding system prompt P_{L_i} .

Tokenization handling is a critical engineering challenge, as non-Latin scripts consume significantly more tokens per semantic unit. To mitigate API limit exhaustion, the system dynamically adjusts the max_output_tokens parameter. Unicode storage handling is strictly enforced as UTF-8 across all Firestore collections.

C. Engagement Score Formulation

To provide caregivers with actionable data, Mitra calculates a daily Engagement Score (ES_d) for the user. The score is formulated as a weighted linear combination of three normalized parameters:

$$ES_d = \alpha \cdot msgs_d + \beta \cdot duration_d + \gamma \cdot reminder_confi$$

Where:

1. $msgs_d$ is the normalized volume of messages exchanged.
2. $duration_d$ is the normalized cumulative time spent interacting with the application modules (Chatbot, Brain Games).
3. $reminder_confirm_pct_d$ is the percentage of scheduled routines explicitly confirmed.
4. α, β, γ are empirically tuned weighting coefficients.

Anomaly Detection:

The system calculates a 14-day rolling average μ_{14} and standard deviation σ_{14} of the Engagement Score. A Z-score is calculated daily:

$$Z_d = \frac{ES_d - \mu_{14}}{\sigma_{14}}$$

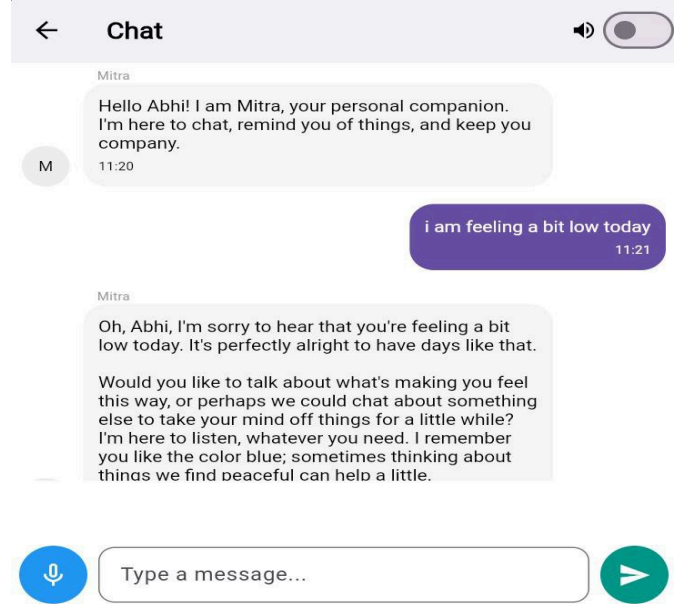
If $Z_d < -2.0$ for two consecutive days, the system triggers an alert to the caregiver dashboard, indicating a potential behavioral deviation.

VI. USER INTERFACE AND FEATURE SETS

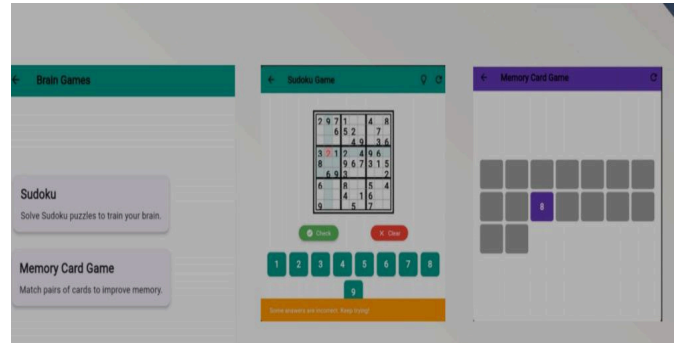
The client-facing architecture of Mitra is fundamentally centered on gerontological design principles. Cognitive load is minimized through a highly restrictive, minimalist navigational paradigm featuring high-contrast palettes and large typography.

A. Elder User Features

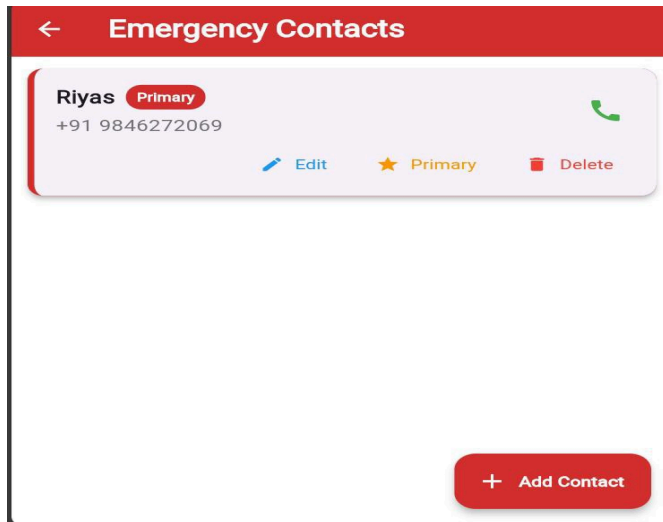
Companion Chatbot: The core feature, powered by the Gemini API, offering voice-first conversational support, emotional check-ins, and contextual memory retrieval.



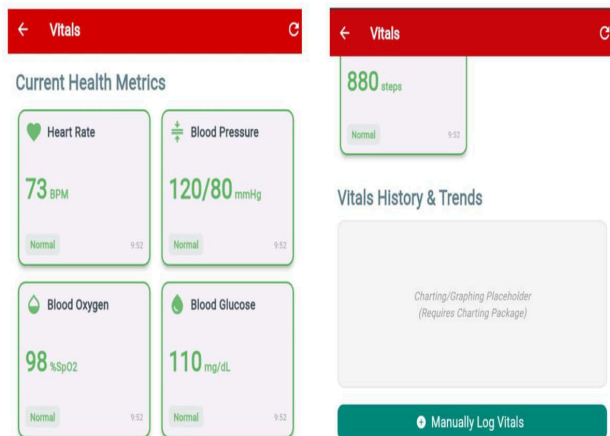
Brain Games: A dedicated cognitive stimulation module featuring interactive games such as Sudoku and Memory Match (Card Game). These exercises are integrated directly into the engagement tracking system.



Safety and Emergency (SOS): A prominent panic button that bypasses conversational AI to immediately dispatch SMS alerts with GPS coordinates to primary emergency contacts.



Health Tracker & Reminders: Interfaces for the manual logging of physiological vitals (Heart Rate, Blood Pressure, SpO2, Blood Glucose) and localized pop-up alerts for medication adherence.



B.Caregiver User Features

Caregiver Tools

Monitoring & Health for The Elder

- Health Vitals Status**
View latest vitals and check-in history for The Elder. >
- Manage Medications**
Set and check adherence for The Elder's reminders. >
- Manage Appointments**
View and set up appointments for The Elder. >
- View Smart Alerts**
Check for missed meds, inactivity warnings, or low battery alerts. >

Caregiver Tools

- Send Custom Reminder**
Push an immediate, custom notification to The Elder's device. >
- Remote SOS Activation**
Emergency: Trigger the SOS protocol on The Elder's device remotely. >

Safety & Emergency

- Update Emergency Contacts**
Manage SOS contacts for The Elder. >
- Medical Documents**
Securely access and store prescriptions or reports. >

Remote Monitoring Dashboard: An analytical control center engineered using Language-Neutral Analytics. Regardless of the elder's interaction language, telemetry data and adherence statuses are aggregated and presented in the caregiver's preferred UI language.

Vitals & Adherence Tracking: Real-time visual representation of the elder's logged health metrics and confirmation status of daily medication routines.

VII. PROTOTYPE IMPLEMENTATION AND VISUAL INTERFACES

To demonstrate the technical viability of the proposed architecture, a fully functional prototype was developed. The visual interfaces were designed adhering strictly to accessibility guidelines, ensuring high legibility and ease of navigation for the target demographic. The following figures illustrate the implemented workflow.

A. Role-Based Onboarding and Profiling

The application initiates with a mandatory role-selection protocol, securely diverging the user experience. An intuitive profile creation interface then captures essential context, including emergency contacts, hobbies, and practical skills, which are fed directly into the LLM's prompt engineering framework.

Figure 3: UI Screenshot - Role Selection Screen (Elder vs. Caregiver)

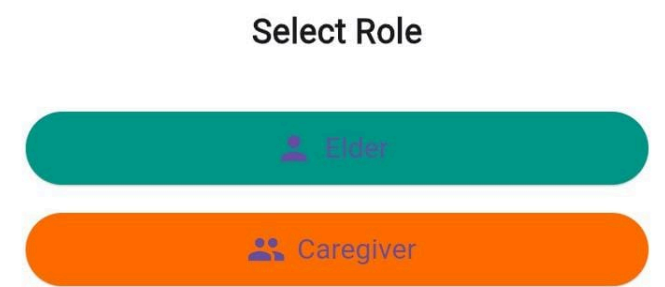
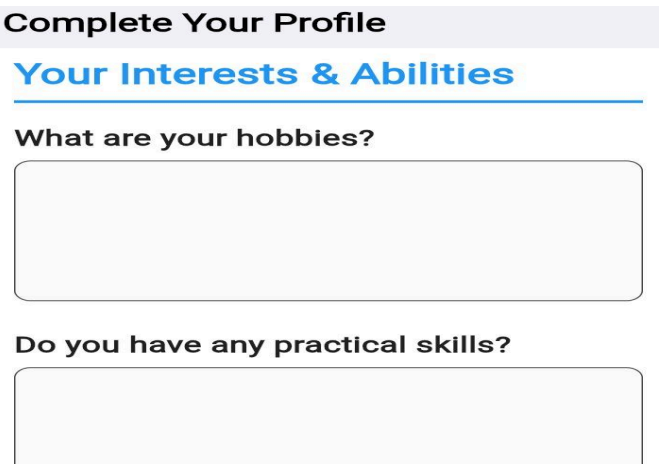
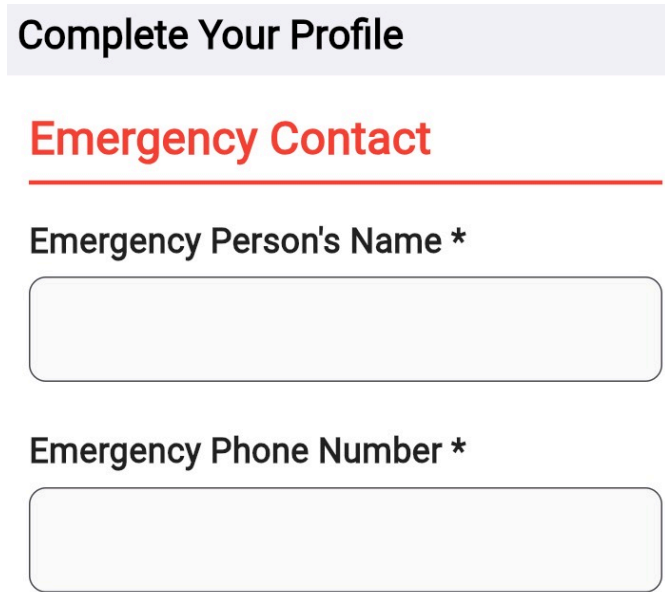


Figure 4: UI Screenshot - User Profiling Interface (Name, Hobbies, Emergency Contacts)



B. The Elder Companion Dashboard

The primary hub for the elderly user is a tile-based dashboard providing immediate access to the conversational agent, cognitive games, and safety tools. The chat interface integrates the localized Gemini API responses with standard messaging UI and voice-input buttons.

Figure 5: UI Screenshot - Elder Main Dashboard Menu

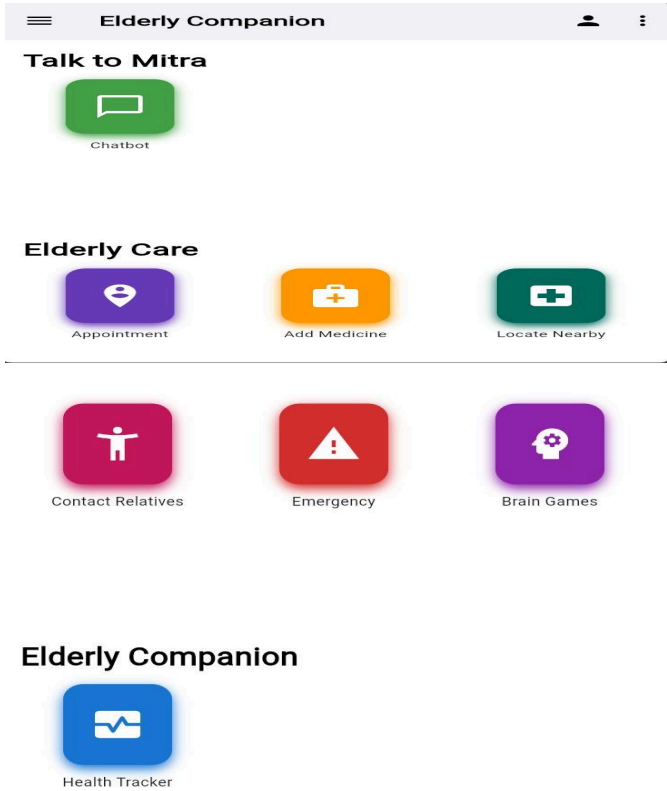
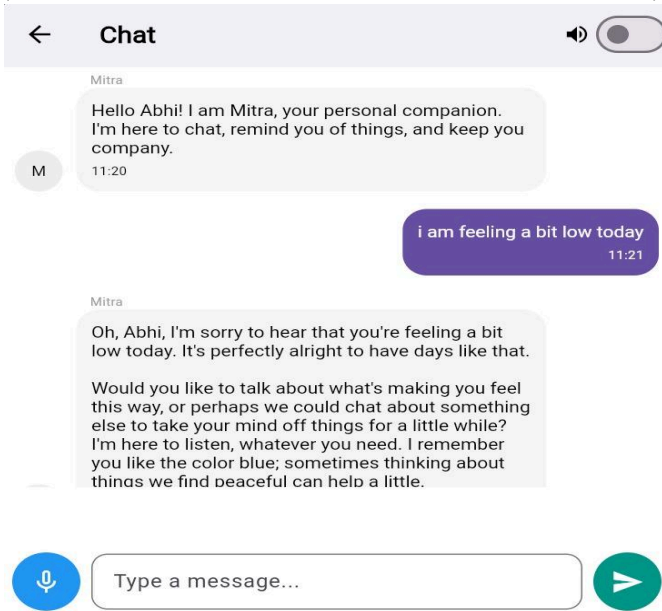


Figure 6: UI Screenshot - Multilingual Chatbot Interface (Active Conversation)



C. Cognitive Stimulation and Safety Tools

The cognitive stimulation module provides interactive mental exercises, while the safety architecture includes vitals logging and a rapid SOS trigger.

Figure 7: UI Screenshot - Brain Games Module (Sudoku and/or Memory Match)

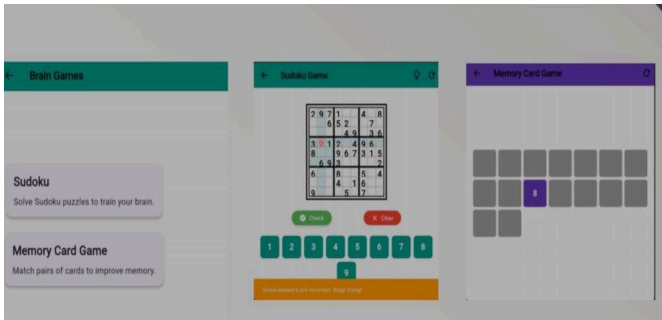


Figure 8: UI Screenshot - Health Tracker/Vitals Logging Interface

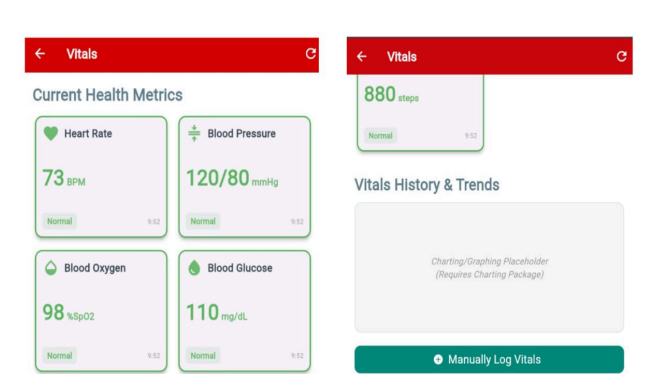
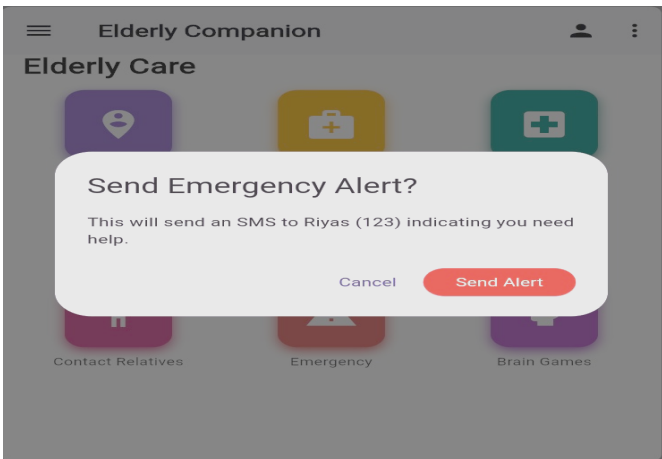


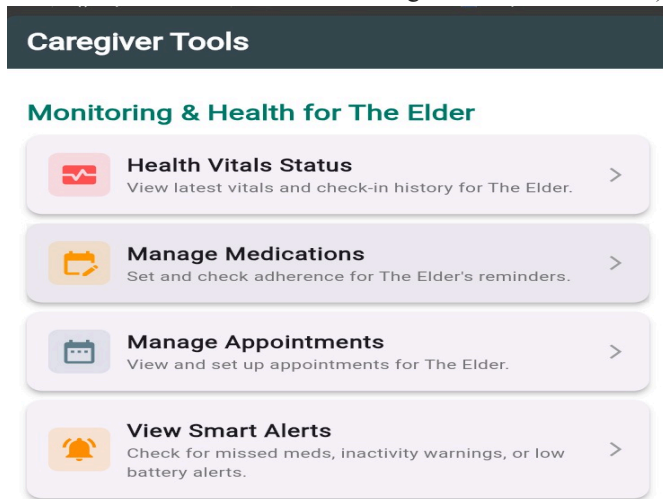
Figure 9: UI Screenshot - Emergency SOS Confirmation Alert



D. Caregiver Monitoring

The Caregiver Dashboard aggregates logged data, presenting it through an analytical interface. This allows proxies to monitor physiological vitals, medication adherence, and algorithmic engagement trends remotely.

Figure 10: UI Screenshot - Caregiver Dashboard (Vitals and Adherence Monitoring View)



VIII. SECURITY AND ETHICAL CONSIDERATIONS

The deployment of AI in vulnerable gerontological populations demands rigorous ethical and security frameworks.

Encryption: Mitra utilizes AES-256 encryption at rest within Firestore. Network transit is secured via TLS 1.3.

Secure Multilingual Storage: All localized data is sanitized to prevent injection attacks specific to diverse unicode encoding standards.

Data Minimization: The database actively truncates raw conversational logs, relying on distilled, anonymized summary strings for long-term context.

Responsible AI Compliance: Prompt engineering constraints explicitly forbid the LLM from issuing medical diagnoses. The multi-shot prompts redirect the model into a supportive posture when confronted with medical queries, actively preventing biased language responses.

IX. LIMITATIONS

While Mitra successfully demonstrates a robust multilingual architecture, several limitations remain.

Token-Length Inefficiencies: Non-Latin scripts exhibit severe token inflation. Over extended conversations, this rapidly approaches the context window limits of the LLM and increases API operational costs.

Model Bias Across Languages: While the model is multilingual, the depth of its cultural alignment may be skewed toward contexts dominant in its pre-training data. Empathetic nuances in complex regional dialects may

occasionally be misinterpreted.

Limited Dialect Support: The current STT and LLM pipeline performs well on standardized macro-languages but struggles with highly localized vernaculars and dynamic code-switching.

Pending Empirical Validation: The current architecture provides a theoretically sound and implemented foundation; however, comprehensive, large-scale clinical trials and performance benchmarking are required to statistically validate its real-world efficacy.

X. FUTURE WORK

Future iterations of the Mitra architecture will focus on rigorous testing and closing the gap between textual processing and multimodal comprehension.

Empirical Evaluation: Conducting comprehensive performance benchmarking (e.g., latency, token overhead) and large-scale user studies to statistically validate the system's efficacy and the accuracy of the anomaly detection models.

Advanced Emotion Detection: Integrating localized sentiment analysis models to adjust the weights of the prompt engineering dynamically based on the user's acoustic stress levels.

Multimodal Voice-Based Input: Transitioning to native speech-to-speech multimodal LLMs to drastically reduce latency and allow the system to detect prosody and tone.

Predictive Cognitive Modeling: Utilizing longitudinal analysis of the Engagement Score to predict early-onset cognitive decline markers.

Integration with Wearable Devices: Ingesting continuous physiological telemetry into the Caregiver Dashboard to supplement the behavioral metrics.

XI. CONCLUSION

This paper presented Mitra, an advanced, multilingual, LLM-driven virtual companion tightly integrated with a clinical caregiver monitoring system. By shifting the paradigm from monolingual, utility-driven assistants to a language-adaptive, empathetic architecture utilizing a hybrid Flutter localization and Gemini LLM API approach, Mitra significantly lowers the barrier to digital adoption for the elderly. The engineering of a translation-independent prompt pipeline, powered securely by Firebase Cloud Functions, combined with the mathematical formulation of behavioral engagement metrics, provides a scalable, dual-purpose solution. It guarantees a hybrid model of independence and supervision, supporting the elderly user while empowering caregivers with proactive health analytics. The architectural design establishes a strong, technically viable foundation for

equitable, globally accessible geriatric care technologies.

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